

Project Documentation

Bank Marketing Campaign Analysis using Excel and Power BI

1. Project Overview

This project focuses on analyzing the **Bank Marketing Dataset** to understand customer characteristics and identify factors influencing term deposit subscriptions. Data preprocessing was performed in Microsoft Excel to clean and prepare the dataset, followed by interactive dashboard creation in Power BI. The project aims to provide meaningful business insights that can help banks improve their marketing strategies and customer targeting.

2. Tools Used

- Microsoft Excel
- Power BI

3. Dataset

Dataset Name: Bank Marketing Dataset

Source: UCI Machine Learning Repository

Key Columns

Age | Job | Marital Status | Education | Account Balance | Housing Loan | Personal Loan | Contact Type | Call Duration | Previous Contact Days (pdays) | Previous Campaign Outcome | Term Deposit Subscription

4. Steps Followed

Data Preprocessing in Microsoft Excel

- Imported the dataset into Excel.
- Checked for missing values and verified data consistency.
- Removed duplicate records.
- Corrected inconsistent values.
- Verified data types for numeric and categorical columns.
- Prepared the cleaned dataset for visualization.

Data Visualization in Power BI

- Imported the cleaned Excel dataset into Power BI.
- Created KPI Cards.

- Designed interactive charts and graphs.
- Added slicers for filtering.
- Implemented Drillthrough functionality for detailed customer information.
- Applied formatting and dashboard layout improvements.

5. Key Insights

- The overall subscription rate is 11.53%, indicating that only a small proportion of customers subscribed to the term deposit campaign.
- Management, Blue-Collar, and Technician are the largest customer groups in the dataset, making them the primary target segments for marketing campaigns.
- Customers aged between 30 and 50 years account for the largest share (63.3%) of the customer base, while customers below 30 years form the smallest segment.
- Secondary education is the most common education level among customers, followed by tertiary education.
- Most customers are married, indicating that married individuals represent the largest customer segment.
- Customers without housing or personal loans form the majority of the customer base, as shown in the loan analysis matrix.
- Cellular communication was used more frequently than telephone for contacting customers, suggesting it was the preferred communication channel for marketing campaigns.
- Previous campaign outcomes show that most customers had no successful previous outcome, highlighting opportunities to improve future campaign effectiveness.

6. Screenshots



Job	Yearly Account balance	Age	Marital Status	Last Contacted Before days	Previous Outcome	Customer Joined	Contact Type	Education	Housing Loan	Credit card
Management	-679	42	Married	343	Failure	No	Cellular	Primary	Yes	No
Management	133	59	Divorced	-1	Unknown	No	Cellular	Primary	Yes	No
Management	252	60	Married	147	Failure	No	Cellular	Primary	Yes	No
Management	255	83	Divorced	-1	Unknown	No	Cellular	Primary	No	No
Management	606	48	Married	-1	Unknown	No	Cellular	Primary	No	No
Management	621	35	Married	208	Failure	No	Cellular	Primary	No	No
Management	695	63	Married	-1	Unknown	No	Cellular	Primary	No	No
Management	1891	54	Married	-1	Unknown	No	Cellular	Primary	Yes	No
Management	2358	59	Married	-1	Unknown	No	Cellular	Primary	Yes	No
Management	2537	48	Married	-1	Unknown	No	Cellular	Primary	No	No
Management	2587	55	Married	84	Success	Yes	Cellular	Primary	No	No
Management	2769	58	Married	-1	Unknown	No	Cellular	Primary	Yes	No
Management	3371	49	Married	-1	Unknown	Yes	Cellular	Primary	No	No
Management	3889	54	Married	-1	Unknown	No	Cellular	Primary	No	No
Management	3953	36	Married	-1	Unknown	No	Cellular	Primary	Yes	No
Management	3983	37	Married	138	Failure	No	Cellular	Primary	Yes	No
Management	4758	44	Married	-1	Unknown	Yes	Cellular	Primary	Yes	No
Management	5310	41	Married	-1	Unknown	Yes	Cellular	Primary	Yes	No
Management	22856	37	Married	888	Failure	No	Cellular	Primary	No	No
Management	-1164	63	Single	-1	Unknown	No	Cellular	Secondary	Yes	Yes
Management	-852	46	Divorced	-1	Unknown	No	Cellular	Secondary	No	No
Management	-665	44	Married	-1	Unknown	No	Cellular	Secondary	No	No
Management	-588	52	Married	-1	Unknown	No	Cellular	Secondary	Yes	No
Management	-538	30	Married	167	Failure	No	Cellular	Secondary	Yes	No

7. Files Included

- **bank.csv** – Original Dataset
- **Cleaned_Bank_Marketing_Data.xlsx** – Cleaned Dataset
- **Bank_Marketing_Campaign_Dashboard.pbix** – Power BI Dashboard
- **Screenshots**
- **Project_Documentation.docx**
- **README.md**

8. How to Use

1. Open **Cleaned_Bank_Marketing_Data.xlsx** to review the cleaned and preprocessed dataset.
2. Open **Bank_Marketing_Campaign_Dashboard.pbix** in Microsoft Power BI Desktop.
3. Use the slicers to filter data by customer attributes.
4. Right-click supported visuals to access the Drillthrough page for detailed customer information.
5. Analyze KPIs and interactive charts to explore customer behavior and marketing campaign performance.